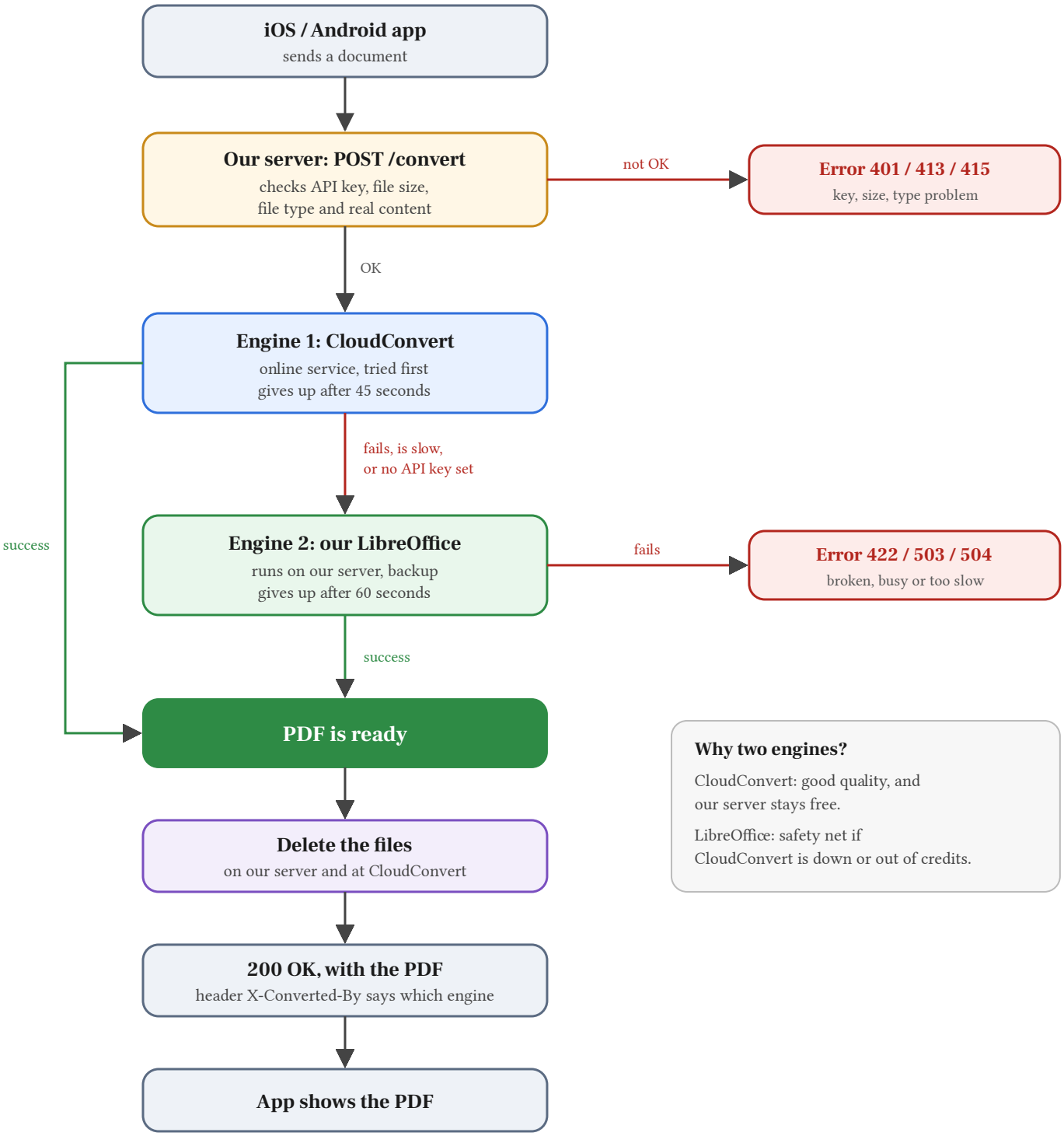
air-printer-converter

How a document becomes a PDF



Who does what

App (iOS / Android)	Picks the file, uploads it, shows the PDF or the error.
Our server	Node + Express. Checks the file, picks the engine, returns the PDF.
CloudConvert	First engine. Online. Costs money per file.
LibreOffice	Backup engine. Free, but uses our server CPU and memory.

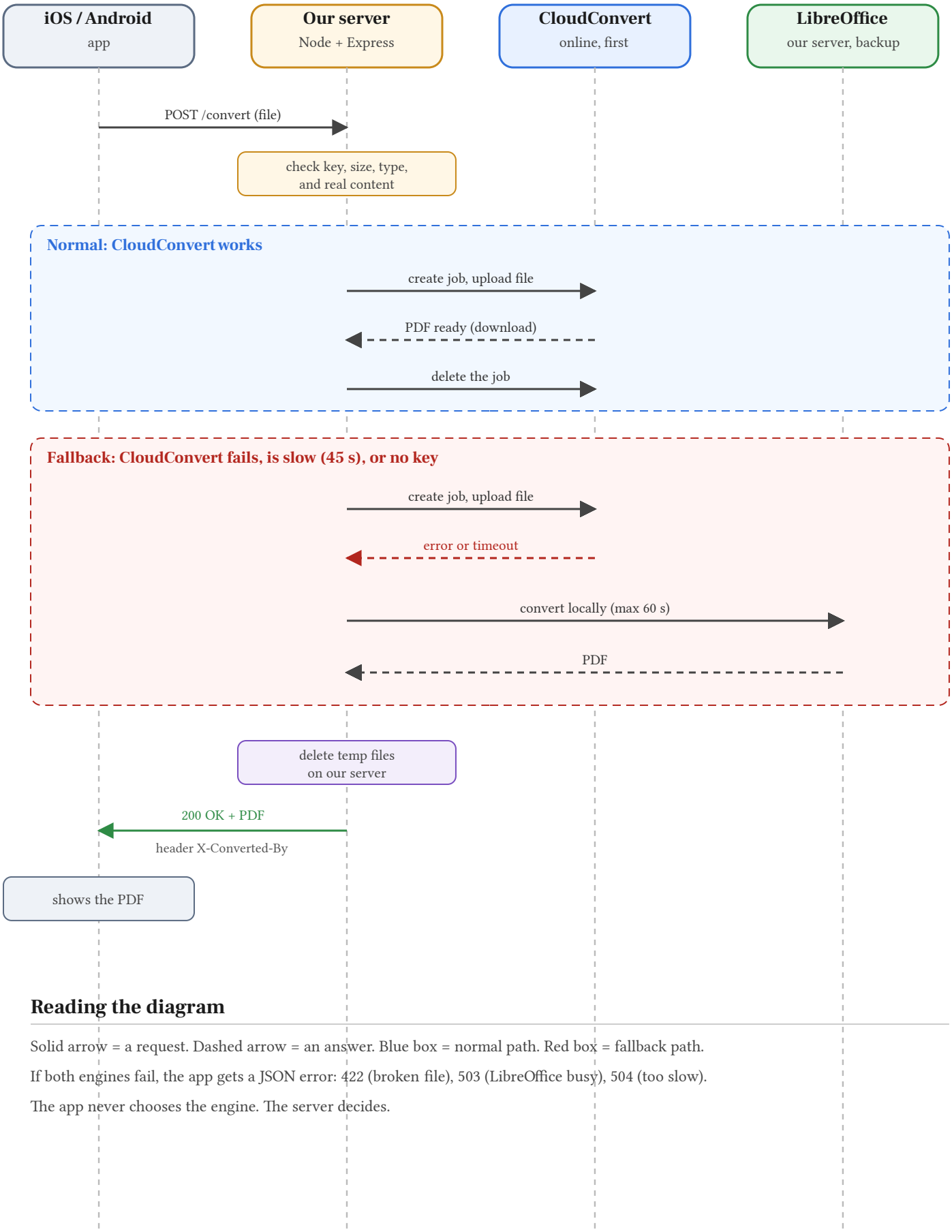
Where the file travels

Phone to our server to CloudConvert and back to our server to phone. All over HTTPS.

If CloudConvert fails, the file stays on our server and LibreOffice converts it.

Request sequence

What happens, in order



Reading the diagram

Solid arrow = a request. Dashed arrow = an answer. Blue box = normal path. Red box = fallback path.

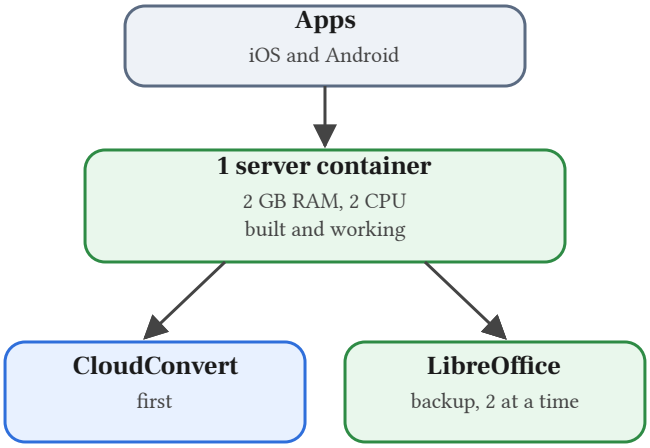
If both engines fail, the app gets a JSON error: 422 (broken file), 503 (LibreOffice busy), 504 (too slow).

The app never chooses the engine. The server decides.

Scaling

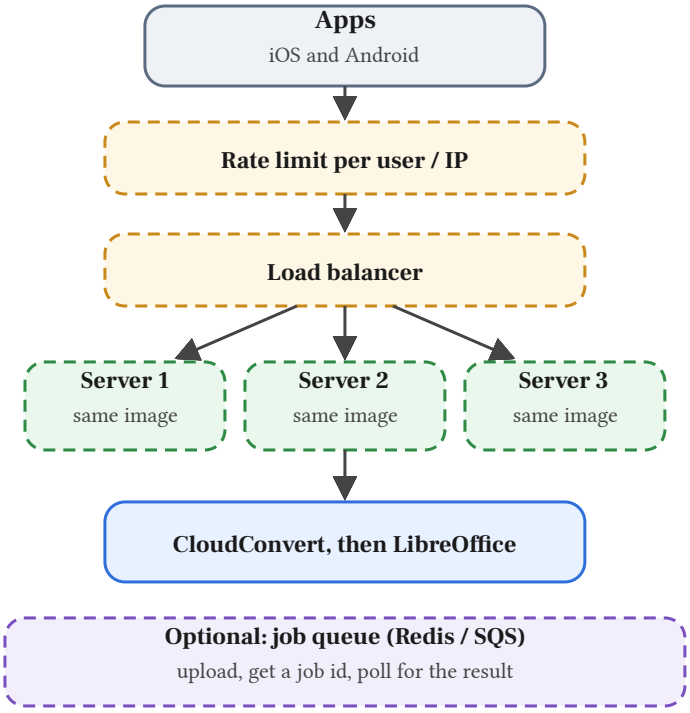
What changes from 100 to 1000 users

Today: about 100 users



Extra jobs beyond 2 at a time get
503 busy, only when CloudConvert fails.

Later: about 1000 users



Dashed = not built yet

What to expect

Users	Peak at once	What you need	Status
~100	1 to 3	1 small server (2 GB RAM, 2 CPU)	Ready
~1000	5 to 20	Several servers, load balancer, rate limit, queue	Not built
any	-	Enough CloudConvert credits (costs money per file)	Your account

Why one server is not enough at 1000 users

Each LibreOffice job takes 5 to 10 seconds and 150 to 500 MB of memory.

The server runs 2 at a time (MAX_CONCURRENT). The rest get 503.

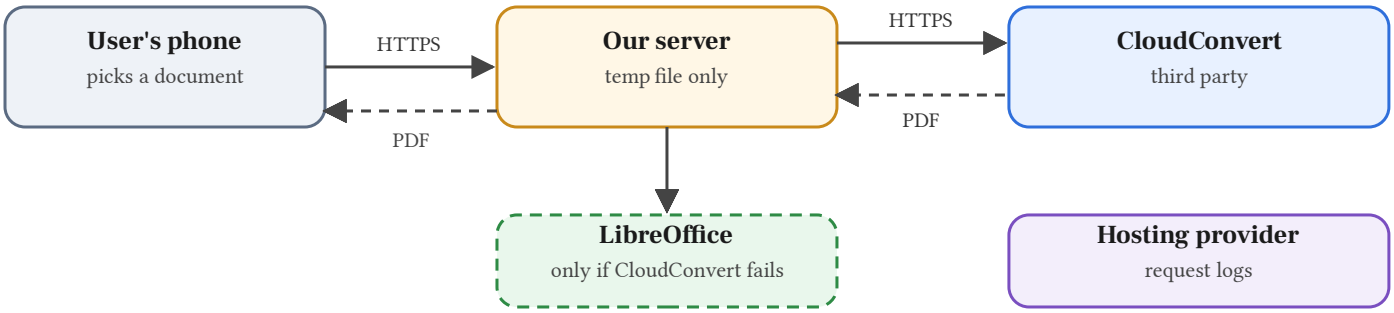
With CloudConvert working, most jobs never touch our CPU. The risk is when it fails or credits run out:

then everything lands on LibreOffice at once.

Fix in order: more servers, then a queue, then a rate limit per user.

Where user data goes

And how long it stays



What is stored where

Place	What	How long
Our server temp folder	the document and the PDF	Seconds. Deleted right after, also on errors.
CloudConvert	the document and the PDF	We delete the job right after. If that call fails, their own retention applies. Read their terms.
Hosting provider logs	IP, time, size, status code	Check your provider. Not the file content.
Our database	nothing	We have no database and no user accounts.

What we promise

- We do not read the documents.
- We do not sell them or use them for ads.
- We do not use them to train AI (our server; see CloudConvert terms for theirs).
- We keep no copy on our server after the conversion.

Must be in the Privacy Policy

- Name CloudConvert as a third party that receives the document.
- Name the hosting provider and say how long logs are kept.
- Tell the user before the first upload: your file is sent to our server and deleted right after.
- App Store privacy form: Other User Content, used for app functionality, not tracking.
- Details: docs/LEGAL_NOTES.md. Have a lawyer check the final text.